

Anish Kumar

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EDUCATION

Indian Institute of Technology(IIT), Patna: CGPA: 7.34 **4 August 2023 – 31 May 2027(expected)**
Bachelor of Technology in Mathematics and Computing *Patna, India*
• Courses: Data Structures and Algorithms(DSA), Machine Learning(ML), Object Oriented Programming, Database Management System (SQL and Relational Database), Probability and Statistics, Numerical Linear Algebra, Discrete Mathematics, Computer Architecture
Senior Secondary (CBSE): Percentage: 91.2% **2023**

PROJECTS

OneIITP (App Development) | | **PlayStore** | | **AppStore** | | **OneAttend** **22 August 2024 – Present**
• **Tech Stack:** React Native, Firebase, JavaScript, Microsoft Azure, Expo, React, Vercel
• Created IIT Patna's official student app serving **2,000+ active users** with features including timetables, attendance tracking, marketplace, and campus resources, increasing engagement by **40%** and achieving **15,000+ reads**, **1,500+ downloads**, and **4.9/5** rating.
• Executed a **QR-based anti-proxy attendance system** with auto-updating codes, reducing proxies by **90%**, and designed **OneAttend** web portal for professors to track and manage attendance details with analytics dashboard integrated with Google Sheets.
• Integrated **Azure authentication** and **Google Sheets API** for real-time bus schedules, improving campus-wide convenience by **30%**.

Facility Routing Engine | | **Github Link** | | **Live Website** **March 2026**
• **Tech Stack:** Next.js, React 19, TypeScript, Zustand, Canvas API, A* / Dijkstra, Vercel
• Built a multi-floor indoor routing platform with a visual map editor for placing rooms, corridors, stairs, elevators, entry/exit points, and connecting them as a weighted graph.
• Implemented emergency-aware pathfinding that supports fire, medical, security, smoke, and hazmat simulations, with node-level incident marking and automatic rerouting around hazardous zones.
• Added route optimization rules for different user roles, including emergency-zone avoidance toggles and fire-safe routing behavior that prefers stairs over lifts during fire scenarios.
• Created reusable scenario presets for complex testing, including a 4-floor hotel layout with 20 rooms per floor, dual lift cores, dual stairwells, and safety nodes such as fire extinguishers.

TPU - Systolic Array MatMul Simulator | | **Github Link** | | **Deployed Link** **April 2025**
• **Tech Stack:** React, JavaScript, Python, Git
• Developed an interactive visualization tool comparing CPU, GPU, and TPU architectures for matrix multiplication.
• Implemented 3 distinct visualization modes (sequential, parallel, systolic array) with step-by-step animations and cycle-by-cycle breakdowns.
• Designed comprehensive comparison dashboard highlighting performance metrics (8 times vs 512 times speed differences) and architectural trade-offs.
• Created responsive UI with interactive matrix editors and real-time visualization of processing element states for educational purposes.

SKILLS

- **Languages:** Python, C, C++, **JavaScript (React/Node)**, **TypeScript**, Wordpress
- **Database:** **SQL (PostgreSQL, MySQL)**, MongoDB
- **Tools/Platforms:** **DSA**, **OOP**, Git, **GitHub**, **Linux**, Windows, **Firebase**, Postman, Vercel, **Docker**, Figma, Gemini API
- **Frameworks:** React.js, Next.js, Node.js, FastAPI, RESTAPI, Tailwind
- **Libraries:** Pandas, NumPy, Random Forest, XGB, **React**, **OpenCV**

ACHIEVEMENTS

- Ranked **186th** globally in **Meta Hacker Cup 2025** (Progressed to Round 3) - **Certificate Link**
- Secured **6th rank** among 6000+ submissions with cash prize in **Volatility Curve Prediction** hackathon on Kaggle by **NK Securities Research in June 2025** - **Certificate Link**
- Finished among **top 10 teams** in **Murf Coding Challenge 2025**, earning certificate and free credits
- **Expert** rated on **Codeforces** (Competitive Programming) - **Codeforces Link**
- Achieved **4 Star** rating on **CodeChef** - **Codechef Link**
- Secured **Global Rank 40** in **CodeChef Starters 219**
- Reached **Round 2** in **The Big Code Challenge by Google 2026**
- Ranked **top 0.76% in JEE Advanced** (1.1M+ candidates)
- Selected for **Summer of Bitcoin 2025**; contributed to **ZeusLN Wallet** in Linux environment under mentorship by implementing and validating on-chain message signing and verification using **yarn run test**, **lint**, **tsc**, and **prettier**.
- Achieved **337** rank among **38,278** participants in **Codeforces Round 1027 (Div. 3)**
- Ranked **under 30** in Bihar Zone in **NTSE Stage 1**

POSITIONS OF RESPONSIBILITY — VOLUNTEER EXPERIENCE

Dev Lead for Cyber Security Problem Statement: *ICTC 3.0* **10 March 2025 - 5 April 2025**
- *Deployed Website Solution* *Problem Statement*
Dev & Design Team: *Inter IIT Cult 2024, IIT Patna* **December 2024**
Overall Tech Coordinator *Anwesha (Cultural Fest of IIT Patna)* **April 2025**

CERTIFICATIONS

Meta Hacker Cup 2025:	Qualified till Round 3  Certificate Link	2026
Certificate of Achievement:	NK Securities Research Hackathon  Certificate Link	9 June 2025
Certificate of Achievement:	Murf AI Coding Challenge 2025 Hackathon  Certificate Link	9 June 2025
Data Structures and Algorithms with C/C++:	Pregrad  Certificate Link	May 2024
Generative AI Study Jams 2024:	Google Developer Groups IIT Patna  Certificate Link	5 August 2024
Google Cloud Computing Foundations:	Infrastructure in Google Cloud  Certificate Link	26 October 2023
Deloitte Data Analytics Job Simulation on Forage	 Certificate Link	6th May 2025